



NANYANG
TECHNOLOGICAL
UNIVERSITY

ML in Banking and Finance: An Introduction

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Outline

- Introduction
- Where is ML/AI used in finance?
- Types of data in Finance
 - Data life cycle
- Pertinent factors for ML/AI in Finance
 - Privacy, Security
- Data governance

Learning Outcomes

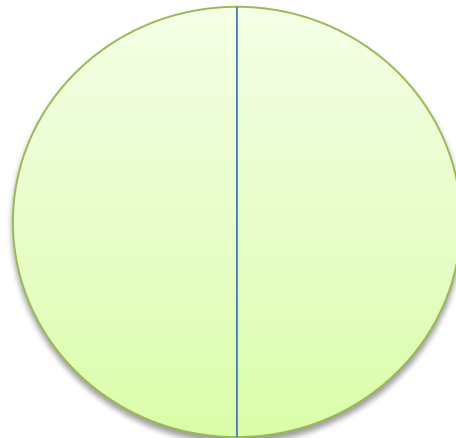
Machine Learning in finance: opportunities and limitations

- Introduction to ML in Finance
- Where is ML used?
 - in banking, news monitoring and customer prospecting
- Some (early) applications of ML in finance

AI in Finance and Banking

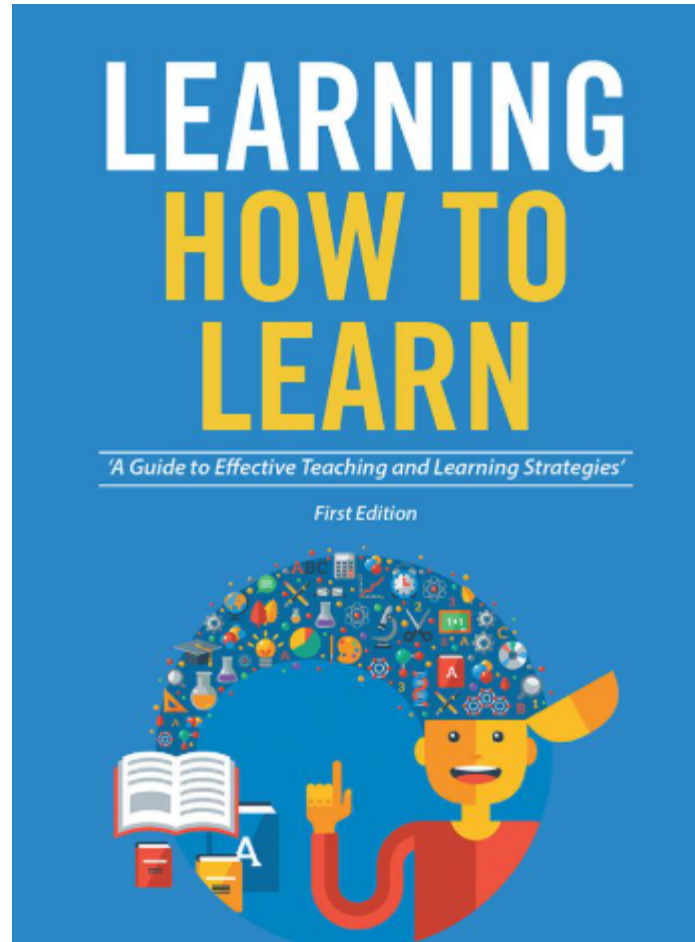
- This course will cover skills/ knowledge in the financial context where AI/ML is applied.
 - For e.g. anti-money laundering/ combat terrorism, technical analysis, portfolio theory.
- Not just the AI/ML methods that are used:
- Successful application of the AI/ML methods requires a **good domain knowledge** of the use case.

Finance
context 50%



AI/ML
methods
50%

How this course will be conducted



Learn how to
learn and not
learn

Rather **apply** and
get things done!
For work

Won't go **overly**
technical

What makes AI/ML in Finance Different?

- Economic Issues:
 - Systemic risks. (got to do with the livelihoods of billions)
 - Human behaviour (not natural laws)
 - Self-learning
 - Unpredictable
 - IMF Study: Potential Benefits
 - Efficiency
- 

Use of ML and Analytics in Finance

- *Financial Operations (General)*
 - *Credit Loans
 - ***GenAI and LLMs**
- Financial Markets
 - For trading and risk management
 - Algorithmic trading
 - Robo advisory
- Consumer banking/ service
 - *Customer prospecting/ segmentation
 - Chatbots
- Regulatory compliance
 - AML/CBT
 - KYC news

Course Plan (Thematic Focus)

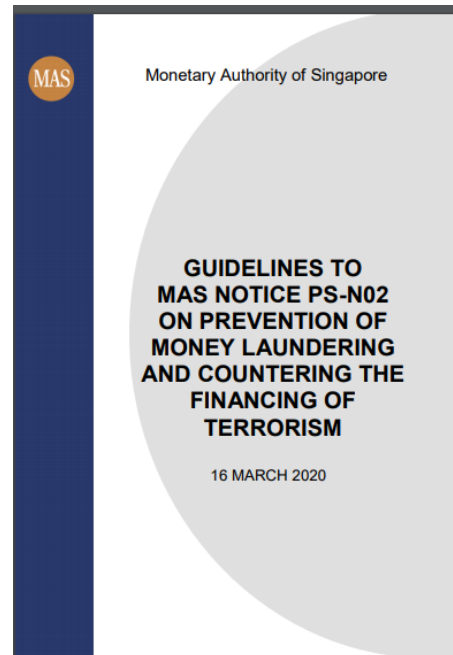
We ek	Topic	Financial Knowledge (Theory)	Ai/ML and Data Component (Practical)
1	Introduction to AI in finance	Data governance, AI explainability	Using Data APIs
2	Financial operation	-	Logistic Regression (binomial), Naïve Bayes, Decision trees Evaluation metrics for supervised learning.
3	Portfolio management	Risks and returns in investing. CAPM, Factor Investing	Boosting methods, ordered logistic regression model

Course Plan (Thematic Focus)

We ek	Topic	Financial Theory	Ai/ML methods
4	Algorithmic Trading	Technical Analysis	Word tokenisation, LLMs
5	Compliance in finance	Regulatory landscape in finance	DBScan/ K-means learning/ Graphical ML
6	Agentic AI	Agent orchestration	

Types of Data in Finance

- Client personal data and transactional data
- Financial data and transactional data
- Textual data
- Risk, control and compliance data



GUIDELINES TO MAS NOTICE PS-N02 ON PREVENTION OF MONEY LAUNDERING AND COUNTERING THE FINANCING OF TERRORISM	
1	Introduction
1-1	These Guidelines provide guidance to all holders of a payment service licence that carry on a business of providing digital payment tokens ("DPT") service (hereinafter "payment service providers") on the requirements in MAS Notice PS-N02 on Prevention of Money Laundering and Countering the Financing of Terrorism – Holders of Payment Service Licence (Digital Payment Token Service) ("the Notice"). These Guidelines should be read in conjunction with the Notice.
1-2	The expressions used in these Guidelines have the same meanings as those found in the Notice, except where expressly defined in these Guidelines or where the context otherwise requires. For the purpose of these Guidelines, a reference to "CDD measures" shall mean the measures as required by paragraphs 6, 7 and 8 of the Notice.
1-3	The degree of observance with these Guidelines by a payment service provider may have an impact on the Authority's overall risk assessment of the payment service provider, including the quality of its board and senior management oversight, governance, internal controls and risk management.
1-4	Key Concepts
	Money Laundering
1-4-1	Money laundering ("ML") is a process intended to mask the benefits derived from criminal conduct so that they appear to have originated from a legitimate source. Singapore's primary legislation to combat ML is the Corruption, Drug Trafficking and Other Serious Crimes (Confiscation of Benefits) Act (Cap. 65A). A payment service provider should refer to the website of the Singapore Police Force's Commercial Affairs Department ("CAD") for more information.
1-4-2	Generally, the process of ML comprises three stages, namely – (a) Placement – The physical or financial disposal of the benefits derived from criminal conduct. Such benefits (or assets) would include digital payment tokens, and other types of virtual assets. (b) Layering – The separation of these benefits from their original source by creating layers of financial transactions designed to disguise the ultimate source and transfer of these benefits. (c) Integration – The provision of apparent legitimacy to the benefits derived from criminal conduct. If the layering process succeeds, the integration schemes place the laundered funds or assets back into the economy so that they re-enter the financial system appearing to be legitimate funds or assets.
1-4-3	As transactions facilitated by payment service providers are unlikely to be predominantly cash based, they are more likely to be used in the layering stage. However, where the transactions involve cash, there is an increased risk of these

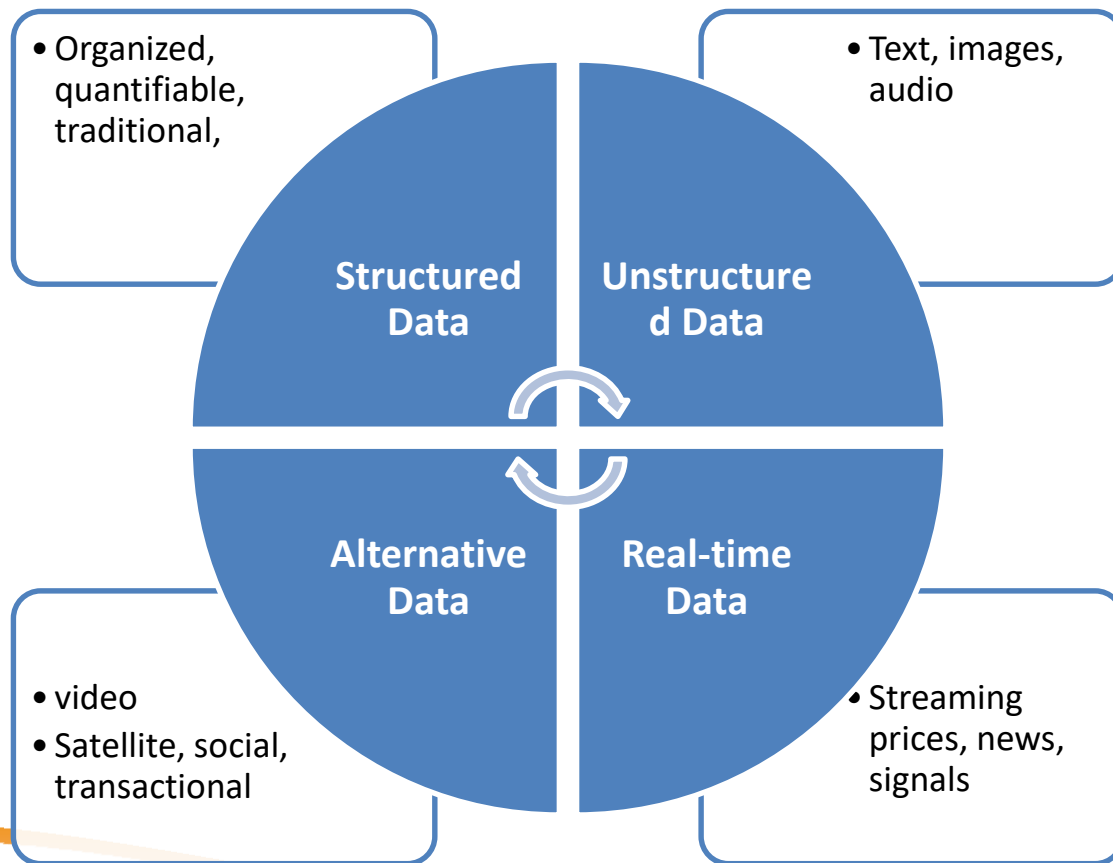
Sources of Financial Data

- Market Data Providers
 - News : from reliable news agencies
 - Refinitiv, Bloomberg
- Exchanges and trading venues
 - TradeWeb, NYSE, SGX etc
- Publicly available sources
 - Company filings (SEC EDGAR, SGXNet, ACRA)
 - Regulatory bodies – FED, World Bank, IMF
- Open source data: web –scraped data, alphavantage, Kaggle, alpaca
- Specialised sources: private company data (PREQIN), pitchbook.
 - BVAL (derived data by Bloomberg)

Which data source is most/ least reliable?

Data Revolution in Finance

- Financial analysis has evolved from purely numerical data to encompass text, images, audio, and alternative data sources. Each type offers unique insights.



Structured Data: Numeric

- Traditional quantitative financial data
- **Price & Market Data**
 - Stock prices, volume, bid-ask spreads, market cap, high/low, open/close. Typically stored in time series format.
- **Financial Statements**
 - Revenue, earnings, cash flow, assets, liabilities, equity. Quarterly and annual reporting from balance sheets, income statements.
- **Financial Ratios**
 - P/E ratio, ROE, debt-to-equity, profit margin, current ratio. Derived metrics for fundamental analysis.
- **Economic Indicators**
 - GDP, inflation, unemployment, interest rates, PMI. Macroeconomic data for top-down analysis.

Data Transformation for Structured Data

Several approaches:-

Smoothing: remove noise from data

Aggregation: summarization, data cube construction

Generalization: concept hierarchy climbing

Normalization: scaled to fall within a small, specified range

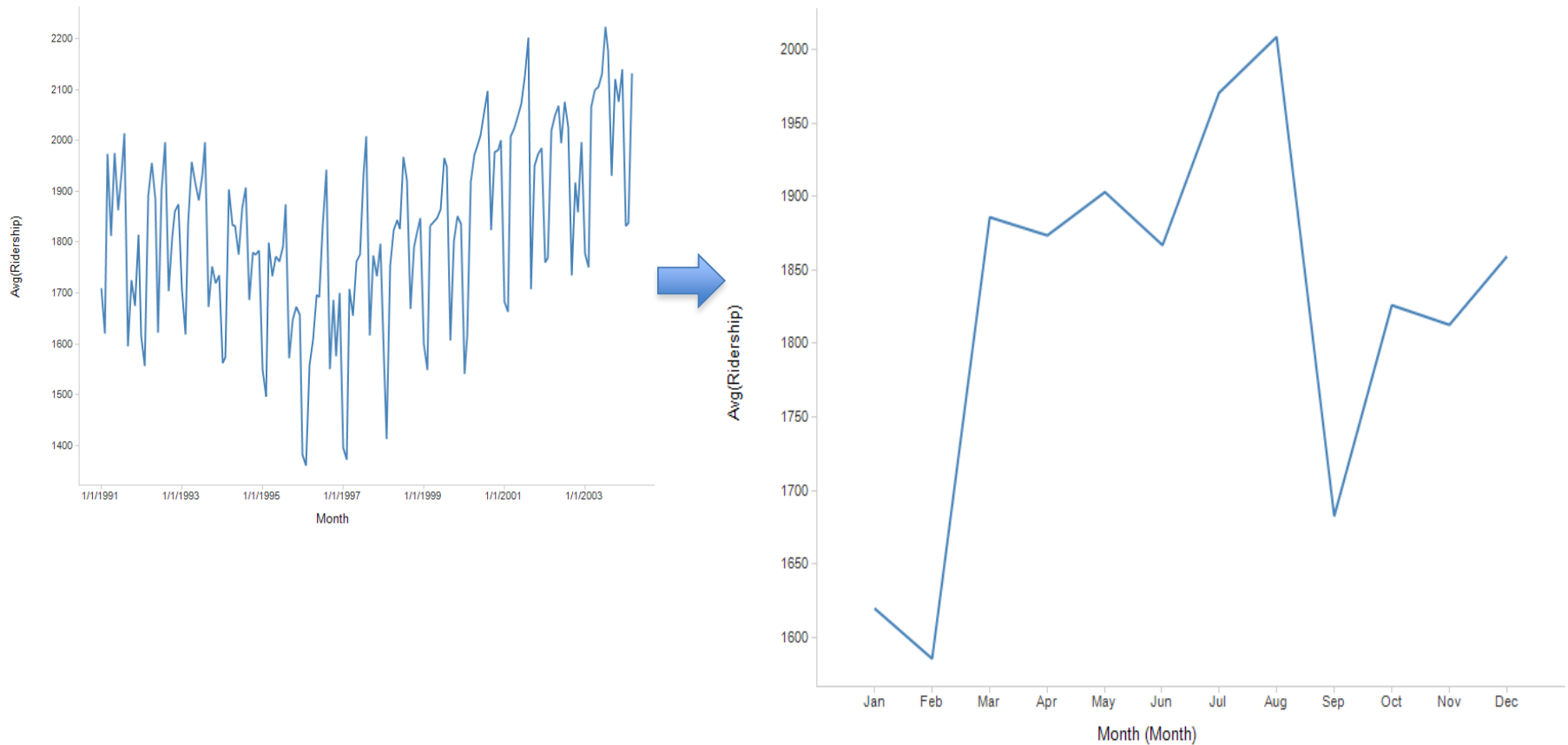
- min-max normalization

- z-score normalization (standardization)

Attribute/feature construction

- New attributes constructed from the given ones

Data Aggregation



Daily values combined to weekly or monthly or yearly

Normalization & standardisation

- **Min-max normalization**: to $[\text{new_min}_A, \text{new_max}_A]$

$$v' = \frac{v - \text{min}_A}{\text{max}_A - \text{min}_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

[Example] Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0].

Then \$73,000 is mapped to $\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$

- **Z-score standardisation** (μ : mean, σ : standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

[Example] Let $\mu = 54,000$, $\sigma = 16,000$. Then \$73,600 is mapped to

$$\frac{73,600 - 54,000}{16,000} = 1.225$$

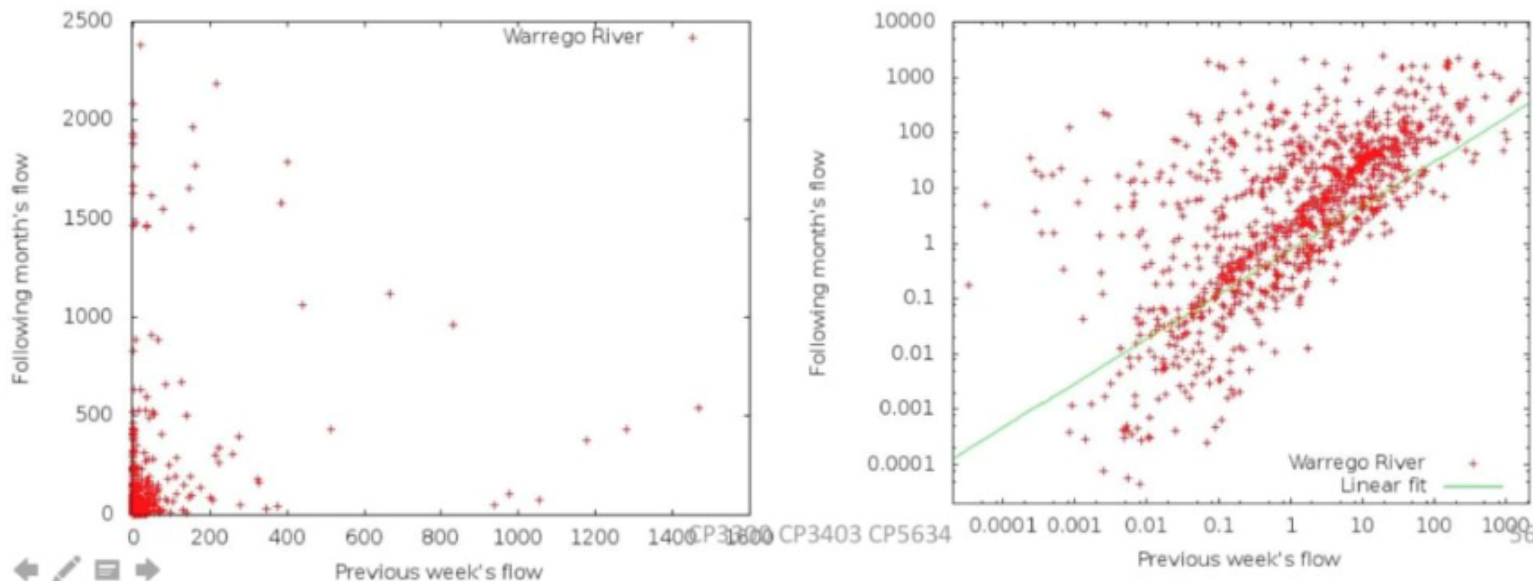
Transformation other techniques

- Decimal Scaling

Range of data can be reduced by dividing with a large number such as 10^x

- Logarithmic Normalisation

Brings down high values, e.g., $\log 10 = 1$, $\log 100 = 2$, $\log 1000 = 3$, etc.



Integration and Transformation

- What are major challenges in **data integration**?
 - Data redundancy
 - Data quality
 - Conflict resolution (we have different values for same data from two different sources)
- What are issues with **data transformation**?
 - Loss of information
 - Subjective sometimes
 - May not help (don't overdo it)

Structured Data: Ordinal & Nominal

- **Ranked by order (Ordinal)**
 - **Credit Ratings**
AAA, AA+, A, BBB, BB, etc. Ordinal scale indicating creditworthiness. Used for bond pricing and risk assessment.
 - **Analyst Recommendations**
Strong Buy, Buy, Hold, Sell, Strong Sell. Consensus ratings aggregate expert opinions into actionable signals.
- **No rank (Nominal)**
 - **Industry Classifications**
GICS sectors, SIC codes, NAICS. Categorical data for sector rotation and peer comparison strategies.
 - **ESG Scores**
Environmental, Social, Governance ratings (A-F or 0-100 scale). Growing importance in sustainable investing.

Ordinal Data: Usage Considerations

Special handling for ranked data

- **Non-Linear Distances**

The difference between AAA and AA is not the same as BB and B. Statistical methods must account for ordinal nature.

- **Encoding Strategies**

Label encoding (1, 2, 3...), one-hot encoding, or learned embeddings. Choice affects model performance in ML applications.

- **Consensus Challenges**

Different rating agencies may disagree. Aggregating ordinal data requires careful methodology (median vs mean).

- **Time Variation**

Ratings change over time. Tracking upgrades and downgrades can signal investment opportunities.

Data Encoding

- Models cannot read in non-numeric data. The unstructured data needs to be discretised.
- Categorical to numerical:
 - Gender (male/ female) -> (0,1)
 - Bank customer type (normal, VIP, priority) - > (1,2,3)
 - Numerical to categorical (age -> below 21 and above 21)
 - Income \$250k -> high income

One-Hot Encoding

- **What is it?**

A technique to convert categorical variables into a binary vector representation where each category becomes a separate column with 1s and 0s.

- **Why use it?**

- Machine learning models require numerical input
- Prevents ordinal relationships between categories
- Each category is treated independently

Example

Original Data:

Color	Value
Red	1
Blue	2
Green	3
Red	4



One-Hot Encoded:

Red	Blue	Green	Value
1	0	0	1
0	1	0	2
0	0	1	3
1	0	0	4

Unstructured Data: Text

The richest source of qualitative information

News Articles & Press Releases

Breaking news drives market reactions. Event studies analyze price movements around announcements. Sentiment analysis extracts tone.

Earnings Call Transcripts

Management tone, word choice, Q&A dynamics. NLP techniques detect deception, confidence, and forward guidance.

SEC Filings (10-K, 10-Q, 8-K)

Risk factors, MD&A sections reveal management concerns. Text analysis tracks changes across filings for early warning signals.

Social Media & Forums

Twitter, Reddit (WallStreetBets), StockTwits. Retail sentiment, meme stock detection, crowd wisdom or madness.

Unstructured Data: Images

Visual data in financial analysis

Satellite Imagery

Count cars in parking lots (retail traffic), oil storage tanks (inventory), construction activity. Alternative data for earnings prediction.

Technical Chart Patterns

CNN models detect head-and-shoulders, flags, triangles. Automate pattern recognition that technical analysts do manually.

Product Images

E-commerce platforms, fashion trends. Computer vision tracks product popularity, pricing strategies, inventory levels.

Document Scans

Historical records, handwritten notes, old financial statements. OCR and image processing digitize legacy data.

Imagery: Satellite data

- Used in crops field identification



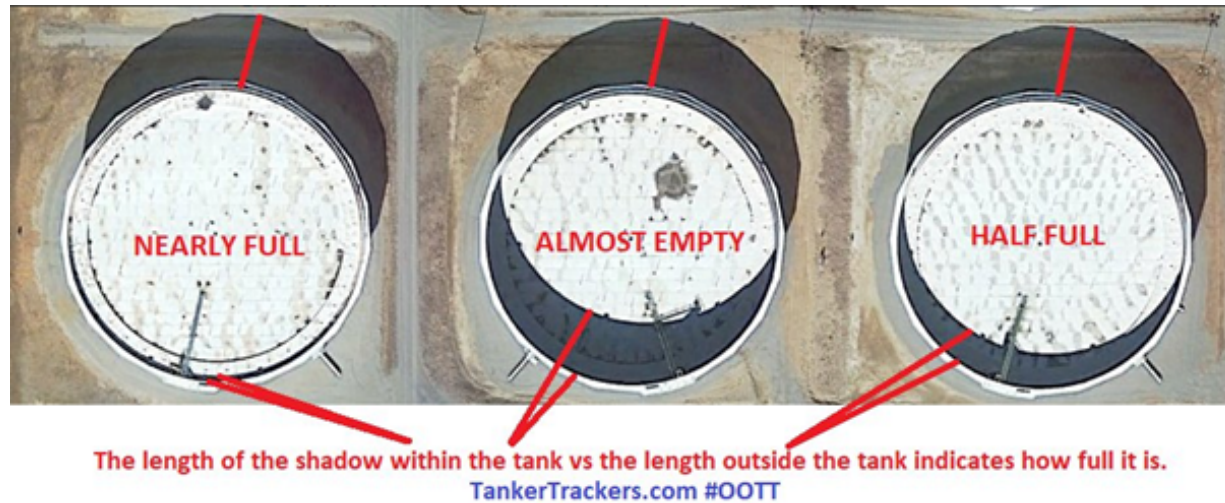
From the satellite imagery, what is the expected harvest (thence supply) of this crop?

- Used in freight movement of crude/ oil for demand/ supply projections

AI in pictures

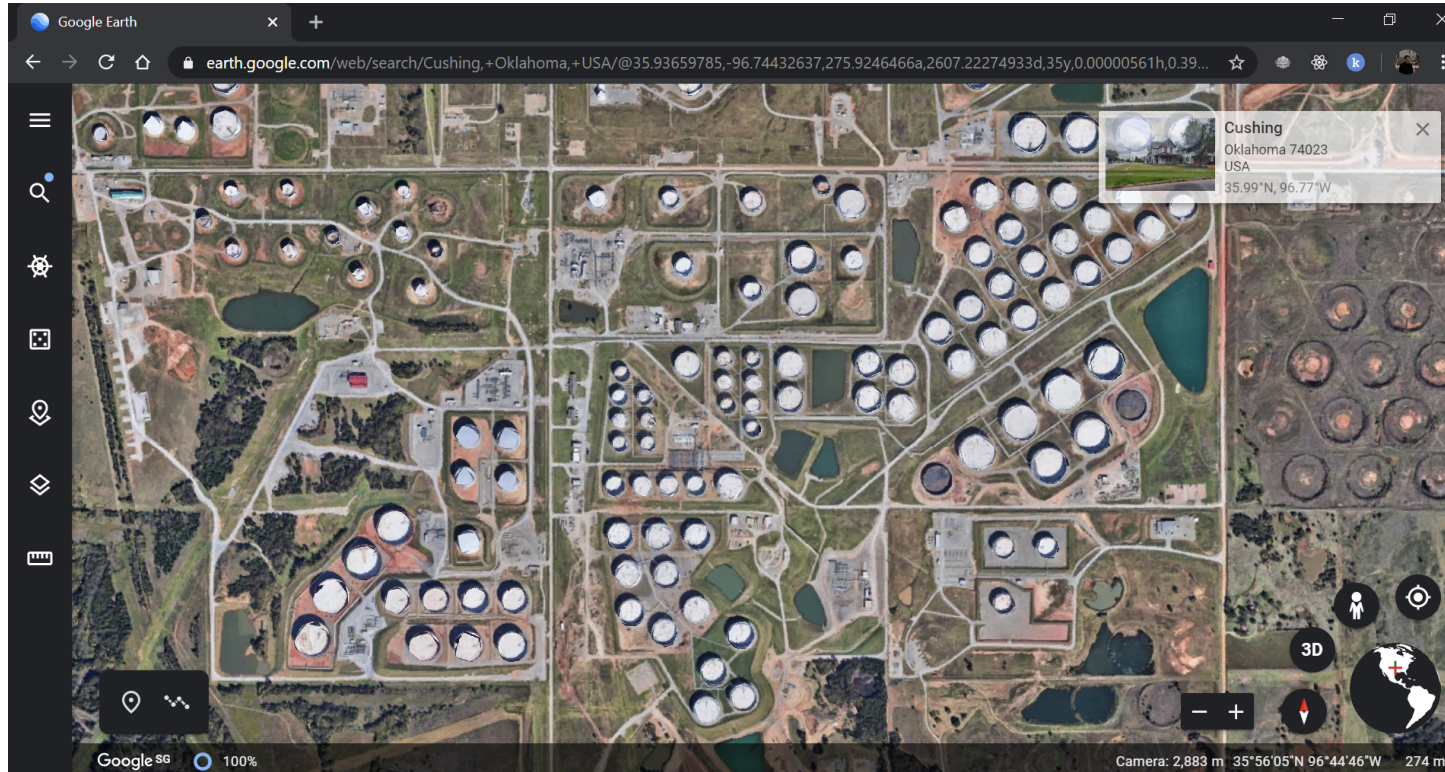
- What happened on 20 Apr 2020?
- Fundamental reason for the negative price – WTI in USA
From the news – poor demand + **lack of storage tankers space**
- AI in pictures!

AI in pictures (satellite)



Generally, the pictures can tell you if an oil storage tank is full or not.

AI in pictures (satellite)



[Google Earth Link](#)

Unstructured Data: Audio

Sound and speech in financial markets

Earnings Call Audio

Voice stress analysis, speech patterns, hesitation markers. Detect CEO confidence or deception beyond words. Tone carries information.

Financial News Podcasts

Transcribe and analyze audio news. Real-time sentiment from CNBC, Bloomberg audio. Faster than text for breaking news.

Trading Floor Audio

Historical: pit trading recordings. Modern: trader communication analysis for compliance, risk management, strategy identification.

Conference Presentations

Investor days, industry conferences. Extract management commentary, Q&A insights, competitive intelligence from audio.

Alternative Data Sources

Beyond traditional financial data

Transaction Data

Credit card purchases, point-of-sale data. Real-time consumer spending trends predict earnings before official reports.

Web Traffic & App Usage

Website visits, app downloads, time spent. Digital engagement metrics forecast revenue for tech companies.

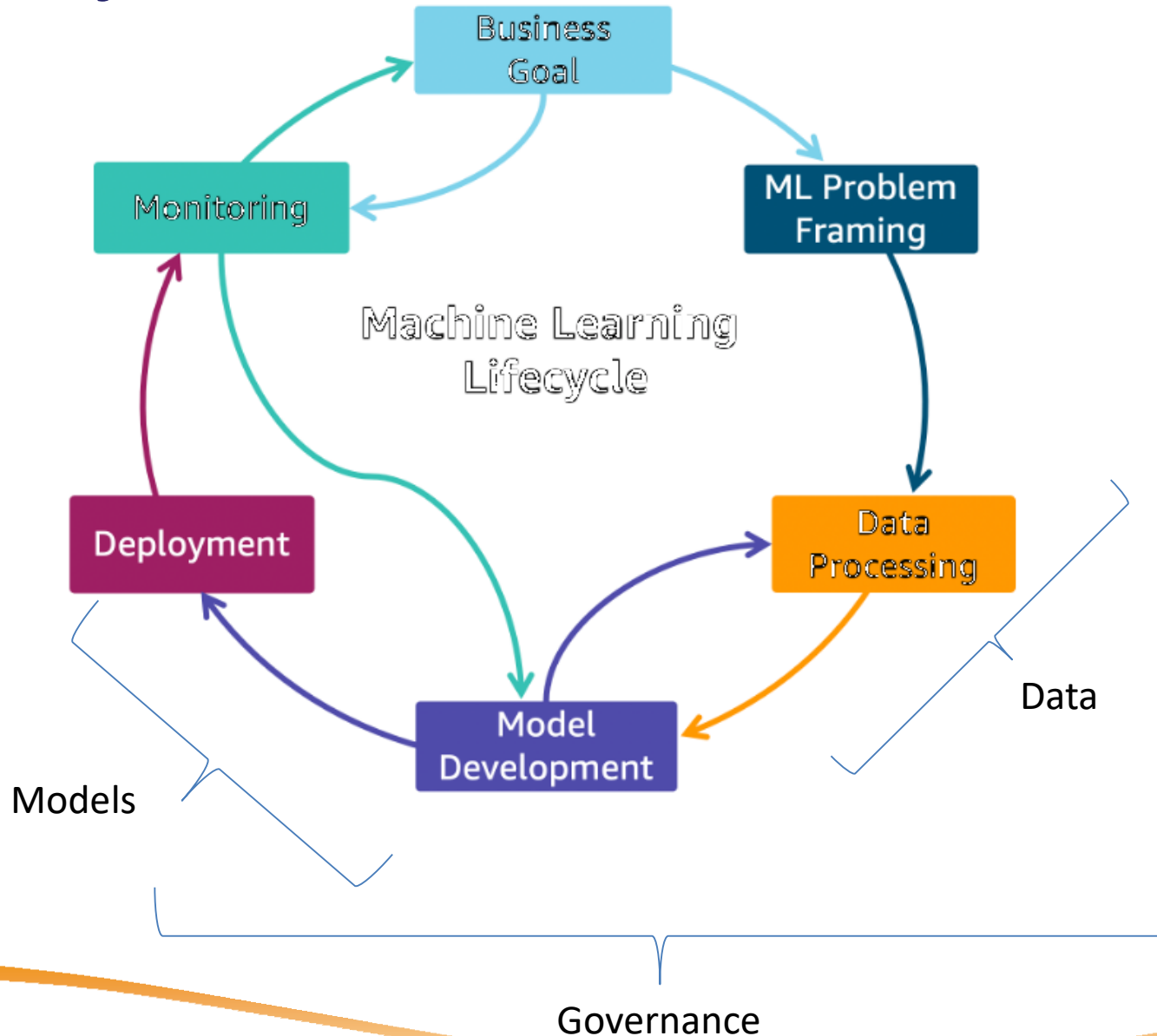
Geolocation Data

Mobile phone location data. Foot traffic to stores, restaurants, attractions. Pandemic-era alternative to traditional metrics.

Weather & Environmental

Temperature, precipitation affect agriculture, energy demand, retail. Climate data for commodity trading and insurance.

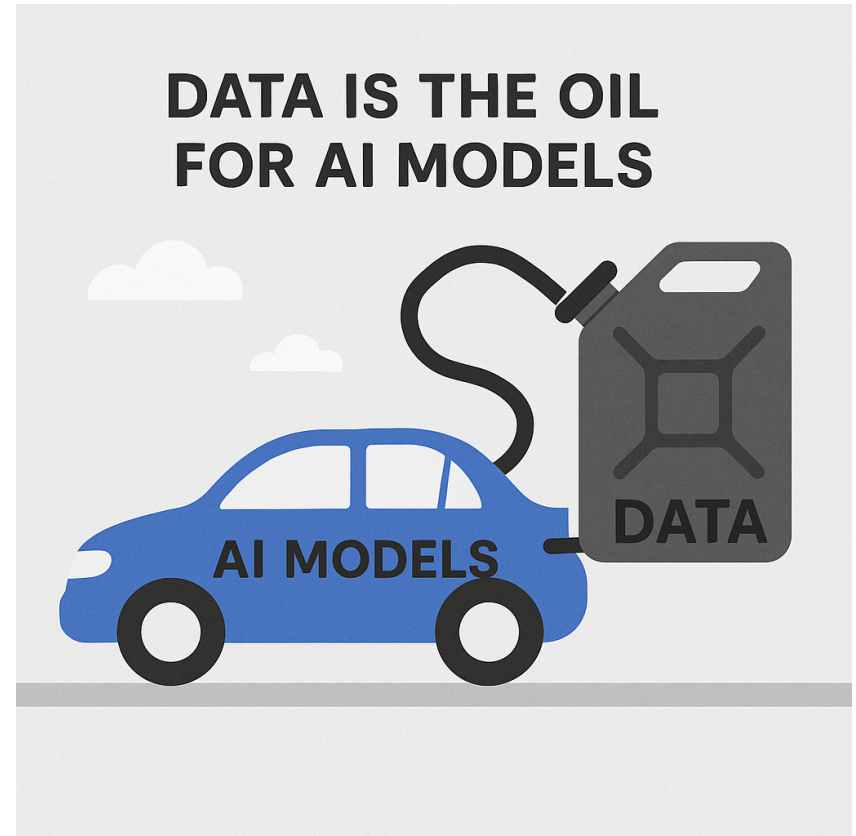
ML Lifecycle



Data and Machine Learning

- Data is the oil that drives the AI/ML engine.
- **Garbage in, garbage out!**

Data is a critical factor for the success of AI/ML projects



Data Integration Challenges

Data Quality & Cleaning

Missing values, outliers, errors. Unstructured data is noisy. Text has typos, images have poor quality, audio has background noise.

Scale & Computational Cost

Processing millions of documents, images, audio files requires significant infrastructure. Cloud computing essential.

Feature Engineering

Converting unstructured data to structured features. Embeddings, sentiment scores, image descriptors must be meaningful for prediction.

Legal & Privacy Concerns

Alternative data may have restrictions. Personal data requires compliance with GDPR, CCPA. Material non-public information risks

MAS AI/ML Governance in Finance

MAS FEAT Principles of AI Governance

1. Fairness

- AI and data-driven decisions should be **accurate, unbiased, and equitable**.
- Customers with similar profiles should be treated consistently.
- Institutions should put in place measures to **identify and mitigate bias** in data, models, and processes.

2. Ethics

AI should be used in a **responsible and transparent way**.

Decisions must align with the **firm's ethical standards and societal norms**.

Institutions are accountable for outcomes from AI use — they cannot “blame the algorithm.”

MAS AI/ML Governance in Finace

MAS FEAT Principles of AI Governance

3. Accountability

Financial institutions remain **responsible for the outcomes** of their AI models. Clear governance structures must be in place, including **board oversight** and **internal accountability mechanisms**.

Human-in-the-loop principles should ensure important decisions are **explainable and reviewable**.

4. Transparency

Institutions should **explain to customers, regulators, and stakeholders** how AI-driven decisions are made.

Transparency should be **proportionate to the impact** (e.g., higher when decisions affect credit, insurance, or investments).

Proper documentation of models, data sources, and processes is essential.

Why AI Governance is important

- Helps ensure **trust in AI adoption** in finance.
- Provides a **balanced framework** — encouraging innovation while managing risks.

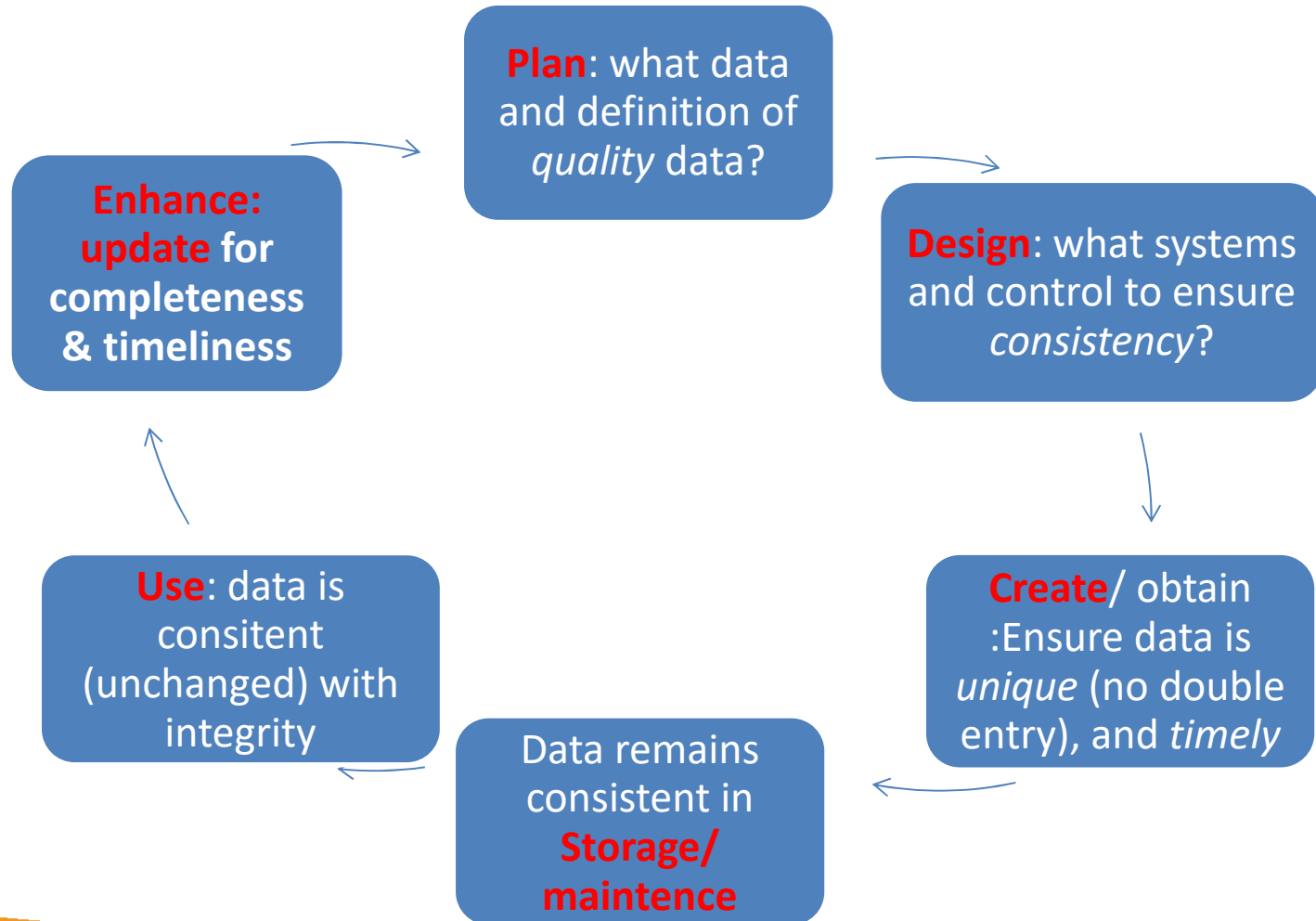
How will AGI impact AI governance?

Explain AGI

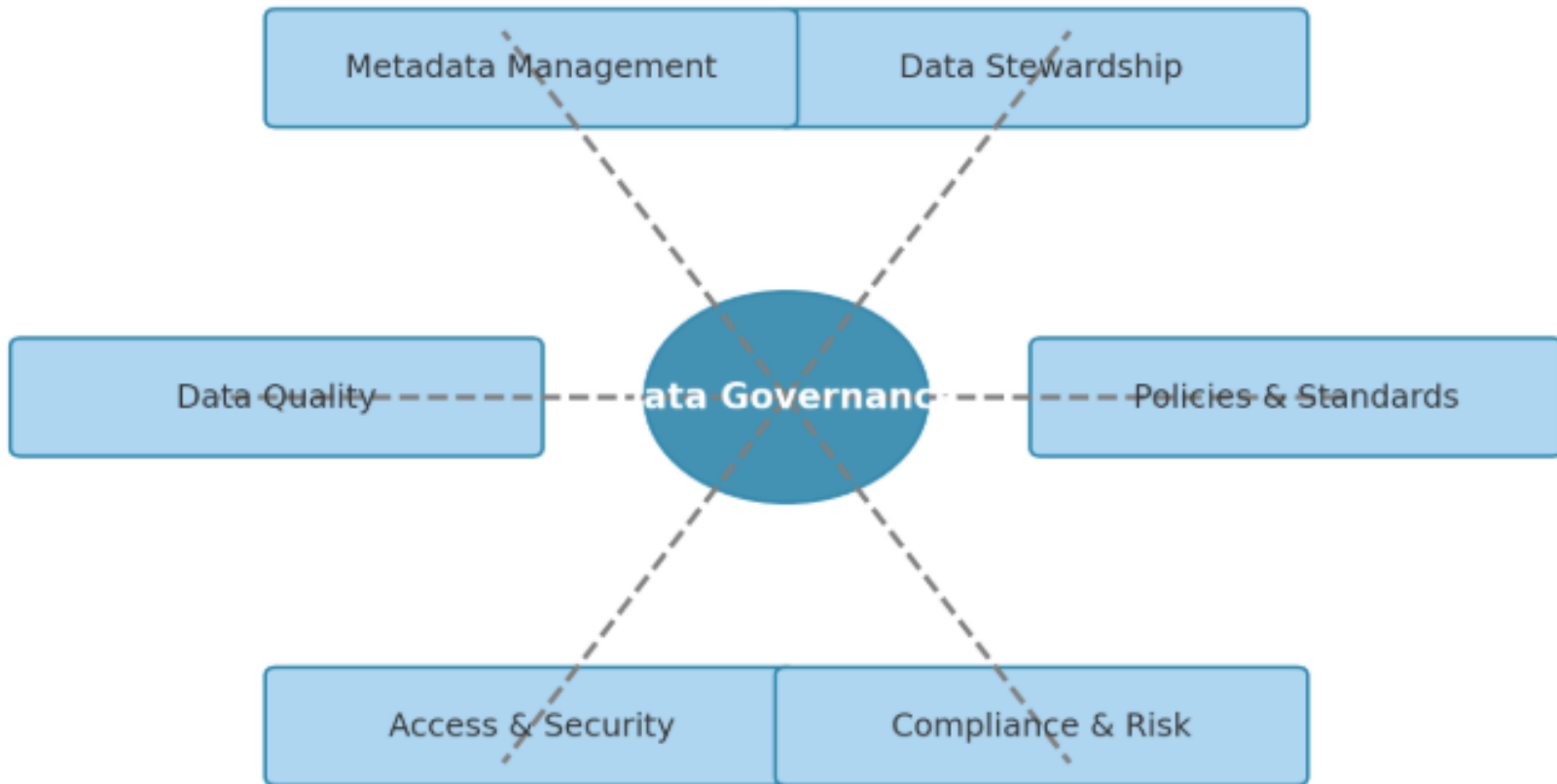
Data Governance framework

- **Data Governance** is the **framework of rules, roles, processes, and technologies** that ensures data is managed as a **trusted, secure, and valuable asset** across an organization.
- **Key Objectives**
 - **Data Quality** – accuracy, completeness, timeliness, consistency.
 - **Data Security & Privacy** – compliance with laws (GDPR, PDPA).
 - **Standardisation** – common data definitions, taxonomies, and metadata.
 - **Accountability** – clear ownership of data assets (data stewards, custodians).
 - **Value Creation** – enable better analytics, decision-making, and innovation.

A Data life cycle



Components in Data Governance



How to implement projects in AI/ML?

- Set your objectives (pick small wins first)
- Decide whether to do inhouse or with outside vendors. What are the pros and cons of this?
- Decide on the time line and the budget.
- Decide what data you need. Do you already have them or need to source for some of them. Are the data confidential and need to satisfy PDPA or other regulations? How should you regulate the data?
- Data pre-processing and cleaning to make it useful for ML algorithms
- Train the AI/ML model. Analyse it for its implications. Run UAT on the model.
- Finally put into operations. Finetune it.

Conclusion

- **Role of AI/ML in Business Finance** – Machine learning and analytics uncover data patterns for predictive and generative tasks across trading, customer prospecting/ segmentation, news monitoring, credit scoring, and regulatory compliance, product recommendation and automation in finance.
- **Project Success & Governance** – Effective AI/ML initiatives depend on clean, well-governed data and adherence to principles (fairness, ethics, accountability, transparency) to build trust and ensure responsible model deployment.

Case Discussion (AI/ML in you workspace)

Discuss the types of projects you encounter among your team.

- A core objective of this course is for participants to identify use cases for AI and/or analytics in their work place. To set the stage:
- Discuss in your groups, the recent AI/ML initiatives that your work place is looking at/ planning to harness this recent advance in technologies.
 - What stage of the project is it at?
 - What technologies are being explored?
 - What are the (operational) objectives?
 - How will the objectives (targets) be measured?
 - What difficulties are being faced by the project implementation?

Please share with the class.

* If there are no plans, consider what type of AI/ML projects you think can benefit your organisation.